

Surname		Group No		Personal Tutor:
First Name		Discipline		

UNIVERSITY OF BIRMINGHAM

School of Engineering

Experiment AM1.2A - Centrifugal Force

Safety

- Normal safety precautions apply (as outlined in the 'Health and Safety – General Guidance booklet).
- Make sure you know where the nearest fire exits are and what the evacuation procedure is
- Maintain a steady footing

Aim

The aim of this experiment is to reinforce your understanding and appreciation of the origin and characteristics of the centrifugal force. The force is not only of fundamental importance, but also lies at the heart of many mechanical devices such as the automatic clutch and can be a source of vibration in rotating machinery which can be destructive.

Introduction

You should be familiar with the standard derivation and simple application of the following centrifugal force equations for a body undergoing circular motion;

$$F = m\omega^2 r = \frac{mv^2}{r}$$

where:

F is the centrifugal force in Newtons

m is the mass of the body in kg

ω is the angular velocity of the body in rad/s,

$v = \omega r$ is the speed of the body in m/s and

r is the radius of the circular motion in m.

In this experiment, when masses are added to a slots in two connected measurement arms and the mechanism rotates (Figure 1), the centrifugal force due the rotating masses transmits to and is measured by the load cell. The load cell measures only this force since the arms themselves are completely balanced. As noted in the introduction out-of-balance forces can be destructive and must be 'balanced' to avoid unnecessary vibration. Therefore to achieve balance a counterbalancing mass is added to the central 'counterbalance arm' which is fixed to the base arm (Figure 2). Masses mounted on this arm counteract the overall imbalanced forces but do not interact with the measurement arms, therefore still allowing the load cell to measure the centrifugal force. Since there are two measurement arms the test masses divide equally between them, to be counterbalanced by a single mass on the centre counterbalance arm.

The base unit has two controls. One adjusts the velocity of the rotating assembly, both in a clockwise direction and an anticlockwise direction. The lights next to this control indicate the direction. The second control is a tare button that allows you to zero the reading from the force sensor before each experiment. The velocity control has three options:

- Normal turning clockwise and anticlockwise to change the velocity with a fine adjustment
- Pressing in while turning to change the velocity with a coarse adjustment
- Pressing in for longer than three seconds to reset the velocity to zero.

You should investigate the centrifugal force by carrying out experiments 1 to 3.

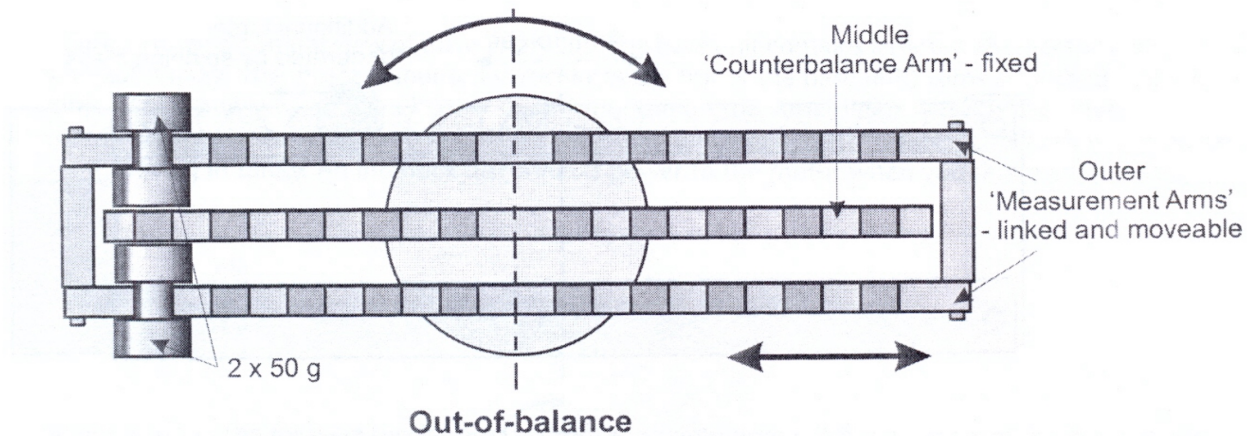


Figure 1: Masses on measurement arms (out-of-balance)

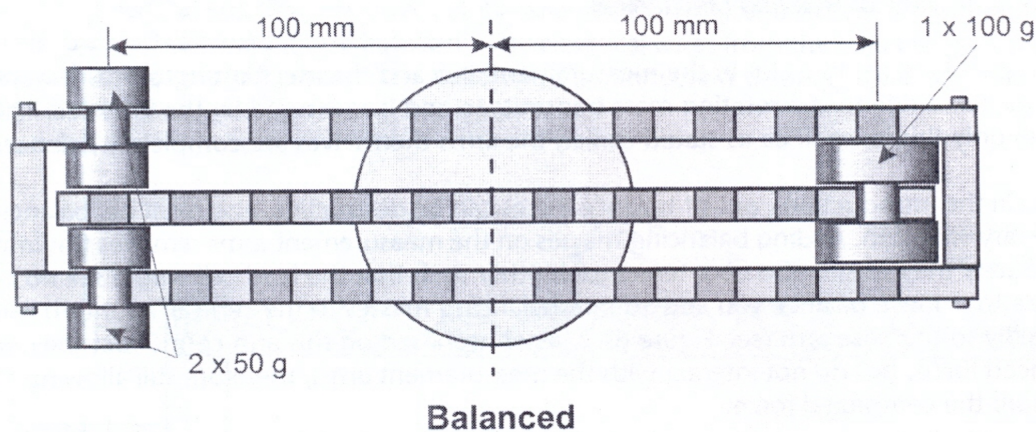


Figure 2: Masses on measurement arms balanced by mass on counterbalance arm

Experiment 1

1. Lift off the safety dome
2. Fit two 50 g matching pair of masses (total 100 g) to the measurement arms at radius 100 mm from the centre
3. Fit a 100 g single mass to the counterbalance arm at 100 mm from the centre (opposite) to balance the two opposite masses
4. Fit the dome and press the button to zero the force reading
5. Set the velocity control (clockwise or anticlockwise) to set the speed to that shown in the first line of table 1.
6. Allow a few seconds for readings to stabilize and record the actual force
7. Repeat for other speeds shown in the table.
8. Calculate the percentage error between the theoretical force and actual force using the equation below:

$$\text{Error (\%)} = 100 \times \left| \left(\frac{\text{Theoretical Force } F - \text{Actual Force } F}{\text{Theoretical Force } F} \right) \right|$$

9. Plot the actual force and theoretical force on the y-axis of the three graphs below.

Radius (r) = 0.1 m; Total mass of matching weights (m) = 0.1 kg					
ω (rad/s)	ω^2	$r\omega^2$	Theoretical Force F (N)	Actual Force F (N)	Error (%)
0	0	0	0	0	0
5					
10					
15					
20					
25					
30					

Table 1: Results of Fixed Mass and Radius, Varied Speed

Experiment 2

Now repeat the experiment at fixed radius of 100 mm and speed of 30 rad/s, varying the total mass of matching weights as in Table 2.

Radius (r) = 0.1 m; Speed (ω) = 30 rad/s; $r\omega^2 = 90$				
Total Mass (g)	Mass m (kg)	Theoretical Force F (N)	Actual Force F (N)	Error (%)
0	0	0	0	0
20	0.02			
40	0.04			
60	0.06			
80	0.08			
100	0.10			

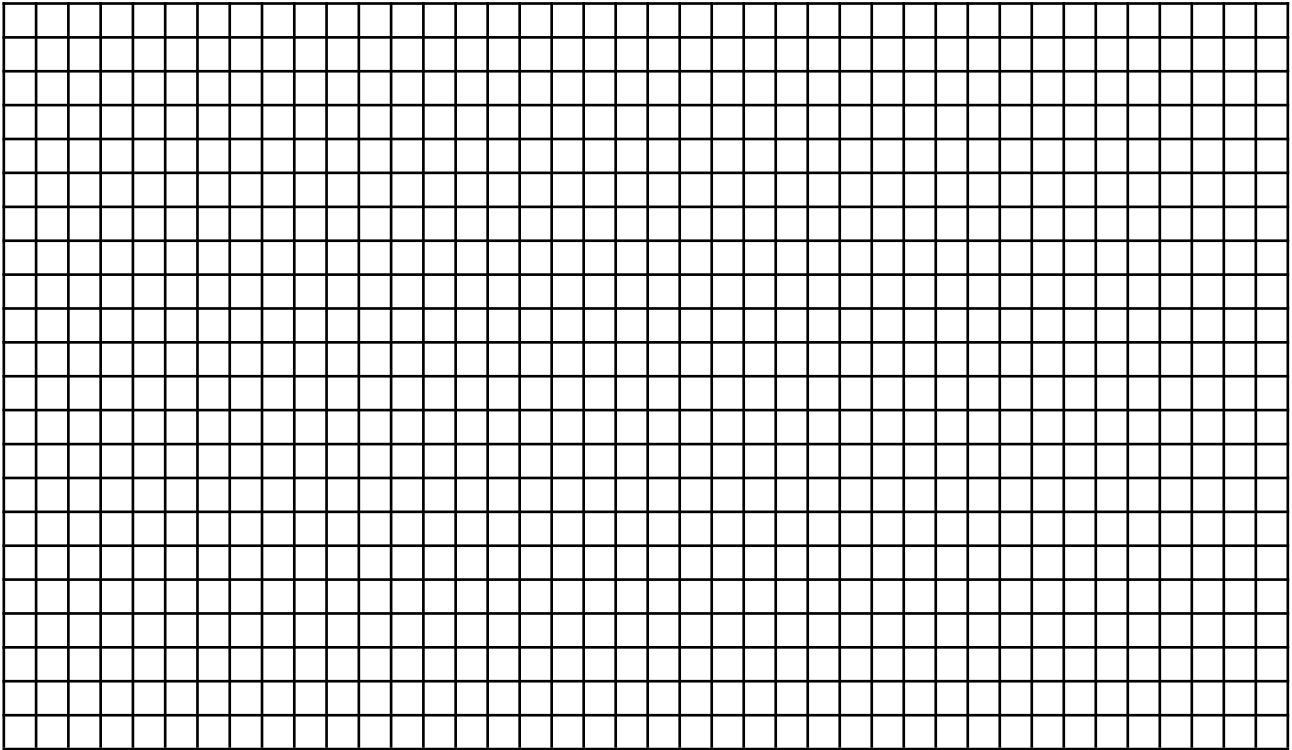
Table 2: Results of Fixed Speed and Radius, Varied Mass

Experiment 3

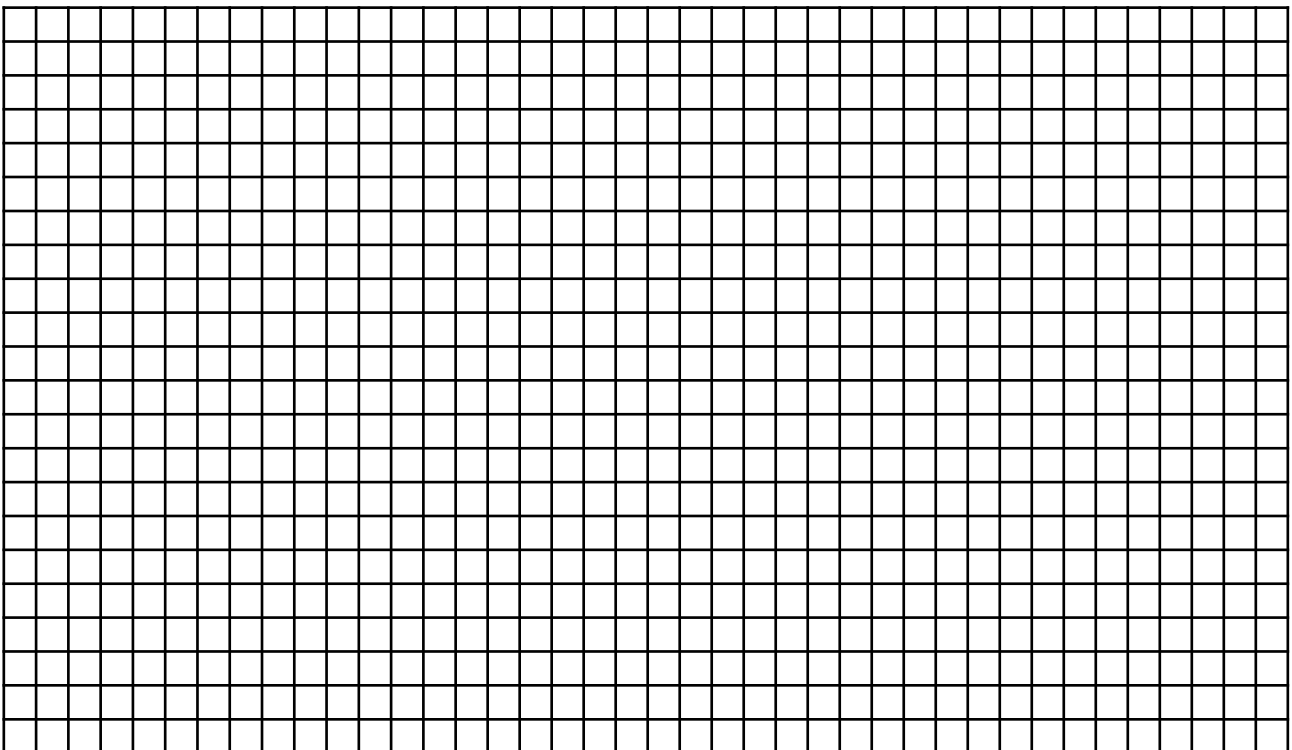
Repeat the experiment at a fixed total mass of 100 g and speed of 30 rad/s, varying the radius at which the masses are placed as shown in Table 3.

Speed (ω) = 30 rad/s; Total mass of matching weights (m) kg = 0.1				
Radius (r) m	$r\omega^2$	Theoretical Force F (N)	Actual Force F (N)	Error (%)
0	0	0	0	0
0.02	18			
0.04	36			
0.06	54			
0.08	72			
0.1	90			

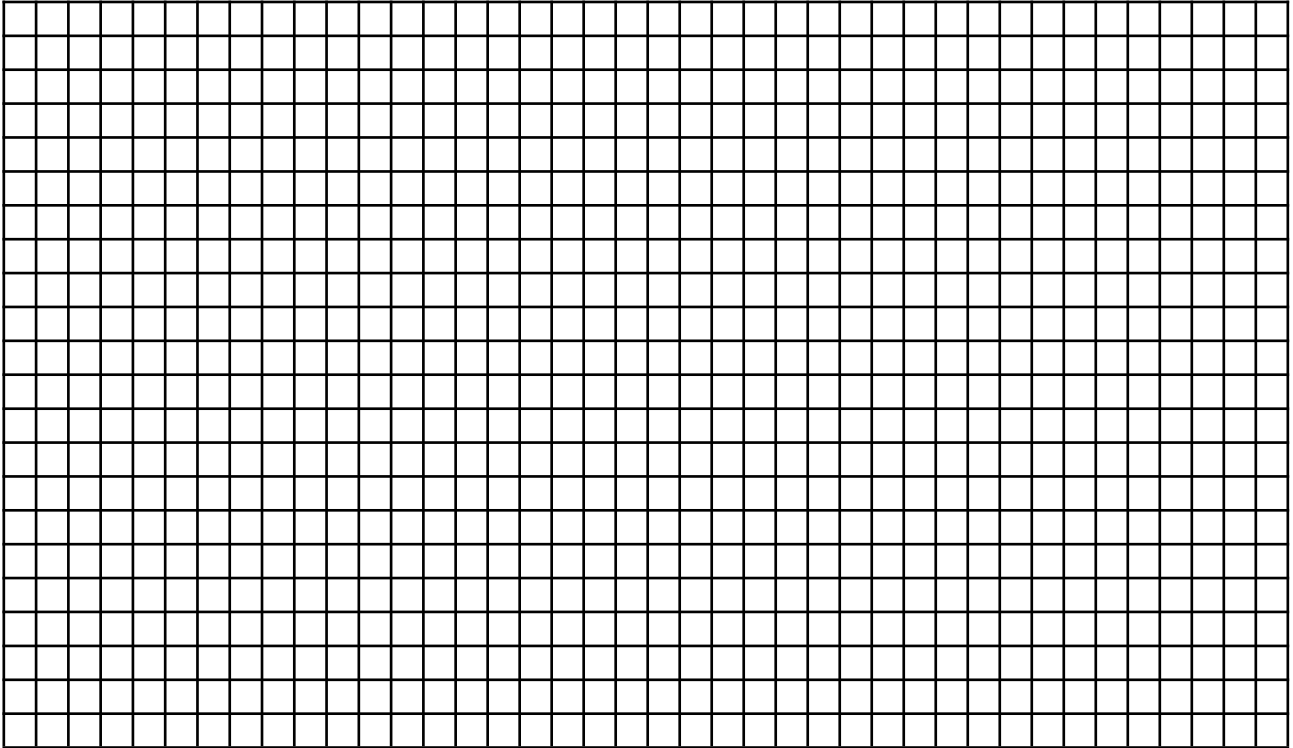
Table 3: Results of Fixed Speed and Mass, Varied Radius



Graph of Centrifugal force vs ω^2



Graph of Centrifugal force vs M_a



Graph of Centrifugal force vs r

Comments on results obtained.